

LiquidPalette

Mobile Web Interface & QR Dispensing Ecosystem for Custom Coffee Kiosks

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1. Executive Summary

LiquidPalette is an interactive mobile web application designed to power the Custom Coffee Color Kiosk ecosystem. Instead of navigating a traditional static menu (e.g., cappuccino, latte, americano), users engage with a tactile, visual "color mixer" interface on their personal smartphones [cite: 1]. By shifting the configuration layer to a QR-redirected mobile web app, LiquidPalette eliminates physical kiosk queuing bottlenecks, supports infinite custom blends, and generates a unique shareable **Coffee Color ID** (e.g., #A96F42) complete with a scannable pickup QR code [cite: 1].

Core Vision: Turn coffee customization from a back-end utility into a delightful, gamified, and highly shareable digital art form where users paint their beverage using a live visual color spectrum.

2. Product Goals & Objectives

- **Zero Kiosk Congestion:** Enable users to browse, customize, pay, and queue their drink asynchronously via personal mobile devices, cutting physical terminal transaction time to under 10 seconds (scanning the pickup QR code).
- **Emotional Engagement:** Transform fluid ratios into a living color swatch that updates in real-time, creating a visceral connection between digital input and physical output [cite: 1].
- **Virality & Retention:** Establish a memorable brand footprint through unique Coffee Color IDs and easy one-tap social sharing or recipe reordering.
- **Hardware Agility:** Decouple the creative experience from the physical dispensing unit, allowing rapid UI iteration without modifying underlying commercial espresso machinery [cite: 1].

3. User Persona & Journey

STEP	USER ACTION	SYSTEM RESPONSE / INTERFACE STATE
1. Discovery	Scans QR code on the physical kiosk or promotional tent card using phone camera.	Instantly loads the lightweight mobile web app without requiring an App Store download or forced account creation.
2. Customization	Interacts with physics-inspired floating controls/sliders for Coffee, Milk, Roast, and Sweetness.	Central liquid swatch fluidly morphs shades and blends colors in real time, displaying exact ratios and live Hex codes [cite: 1].
3. Identity	Views the generated recipe breakdown and unique Coffee Color ID badge.	Assigns a poetic color name and hex value (e.g., #A96F42) with a one-tap 'Save to Favorites' option [cite: 1].
4. Checkout	Reviews pricing, confirms recipe, and completes payment via Apple Pay, Google Pay, or UPI.	Generates a dynamic secure pickup pass featuring a scannable QR code and estimated preparation countdown timer.
5. Fulfillment	Holds phone QR code up to the kiosk scanner; watches automated dispensing.	Machine dispenses exact recipe into a branded cup with a transparent viewing window to verify the final color [cite: 1].

4. Functional Requirements

4.1. The Mixing Playground (UI & Physics)

- **Four Primary Controls:**
 1. *COFFEE / ESPRESSO* (Light → Strong) [cite: 1]
 2. *MILK* (Black → Creamy) [cite: 1]
 3. *ROAST* (Light → Dark) [cite: 1]
 4. *SWEETNESS* (None → Sweet) [cite: 1]
- **Weightless Physics Interaction:** Implement lightweight canvas/physics particle motion or smooth spring animations (via Framer Motion) so controls feel organic and tactile.
- **Real-Time Color Blending Engine:** A central SVG/Canvas liquid element dynamically calculates RGB and Hex values based on slider weighting, giving immediate visual feedback [cite: 1].

4.2. Recipe Engine & Coffee Color ID

- **Ratio Calculation:** Automatically compute precise volumetric percentages (e.g., 62% coffee, 38% milk, Medium roast, 0g sugar) [cite: 1].
- **Color ID Generation:** Map specific hexadecimal codes to each cluster of parameters, rendering a badge that users can copy, share, or save [cite: 1].

4.3. Checkout & QR Pass System

- **Frictionless Payment:** Integrate standard mobile-first payment gateways with one-tap digital wallet support.
- **Secure Tokenization:** Generate a time-expiring JWT-backed QR code payload containing the exact recipe array, sent directly to the dispensing kiosk upon scan.

5. Non-Functional Requirements & Design System

Visual Aesthetic

Neo-brutalist and glassmorphic hybrid. Deep dark charcoal backgrounds (#121212) paired with high-contrast volt-green or neon amber accents, frosted glass containers (`backdrop-filter: blur`), and bold typography [cite: 1].

Performance & Tech Stack

Built with React, Tailwind CSS, and lightweight animation libraries. Optimized for mobile web browsers to guarantee buttery-smooth 60fps rendering across iOS and Android viewports.

6. Success Metrics & KPIs

- **Order Completion Rate:** Target >75% conversion from QR landing to successful payment.
- **Throughput Speed:** Average mobile ordering time under 45 seconds; physical kiosk dispensing scan time under 5 seconds.
- **Viral Coefficient:** Percentage of users who tap the 'Share Color ID' or save recipe button to social channels.
- **Repeat Rate:** Percentage of users returning to reorder a saved Coffee Color ID within 14 days.